Major Project Report

on

Bio-Mimetic Robot

Submitted by

Mihal Palakkal (20223071)
Ruvais P (20223087)
Mohamed Jaseeb P J (20223120)
Rose Maria (20223085)

In partial fulfilment of the requirements for the award of degree of Bachelor of Technology in Computer Science and Engineering.



DIVISION OF COMPUTER SCIENCE AND ENGINEERING
SCHOOL OF ENGINEERING
COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY

OCTOBER 2026

DIVISION OF COMPUTER SCIENCE AND ENGINEERING
SCHOOL OF ENGINEERING
COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY

CERTIFICATE

Certified that this is a Major Project Report titled

Bio-Mimetic Robot

Submitted by

Mihal Palakkal (20223071)
Ruvais P (20223087)
Mohamed Jaseeb P J (20223120)
Rose Maria (20223085)

of VII Semester, Computer Science and Engineering in the year 2026 in partial fulfillment requirements for the award of degree of Bachelor of Technology in Computer Science and Engineering of Cochin University of Science and Technology.

Mr. Damodaran V	Dr. Sheena S	Ms. Amrutha S Nair
Head of Division	Project Coordinator	Project Guide

DECLARATION

We, Mihal Palakkal, Ruvais P, Mohamed Jaseeb PJ, Rose Maria, hereby declare that the project titled “Bio-Mimetic Robot” is a bona fide work carried out by us during the year 2026, under the guidance of Ms. Amrutha S Nair,Department of Computer Science and Engineering,SOE,CUSAT. We further declare that this project has not been submitted, either in part or full, for the award of any degree, diploma, associateship, fellowship, or any other academic recognition in any university or institution.

ABSTRACT

Contents

DECLARATION i

ABSTRACT ii

List of Figures iv

List of Tables v

1 Introduction (Background and Motivation for the Work) 1

2 Problem Statement 2

3 Literature Review 3

4 Dataset Description and Preprocessing 4

5 System Workflow 5

6 Modular Structure 6

7 Tools Used 7

8 Conclusion 8

References 9

List of Figures

Figure No. Figure Name Page No.

List of Tables

Table No. Table Name Page No.

Chapter 1

Introduction (Background and Motivation for the Work)

Chapter 2

Problem Statement

Chapter 3

Literature Review

Chapter 4

Dataset Description and Preprocessing

Chapter 5

System Workflow

Chapter 6

Modular Structure

Chapter 7

Tools Used

Chapter 8

Conclusion

References